Expanding the squared sum we see that there are (i) n squares J_k^2 plus (ii) products of the form $\mathrm{J}_k\mathrm{J}_l$, $k \neq l$. Clearly the former are always one and the latter have zero expectation, thus

$$\mathbb{E}\mathrm{X}_n^2 = n\delta x^2.$$

The variance therefore grows linearly with the number of steps taken.

Now let us take a slightly different perspective. We start with a time interval $[0, \tau]$ and subdivide this into ν steps of size $\delta t = \tau/\nu$. Let $t_n = n\delta t$, $n = 0, 1, \ldots, \nu$ and define

$$\mathrm{Y}(t_{k+1}) = \mathrm{Y}(t_k) + \sqrt{\delta t}\mathrm{J}_k, \qquad k = 0, 1, \ldots, \nu - 1,$$

where $\mathrm{Y}(0) = 0$. Effectively we are making the size of the jump δx taken at each step of the random walk equal to $\sqrt{\delta t}$.

We will use the shorthand $\mathrm{Y}_k = \mathrm{Y}(t_k)$ and $\Delta\mathrm{Y}_k = \mathrm{Y}_{k+1} - \mathrm{Y}_k = \sqrt{\delta t}\mathrm{J}_k$.

The increments are clearly independent. Let us observe the following facts, for $0 \leq k \leq l \leq \nu$

(i) $\mathbb{E}(\mathrm{Y}_l - \mathrm{Y}_k) = 0$,
(ii) $\mathrm{Var}(\mathrm{Y}_l - \mathrm{Y}_k) = t_l - t_k$.

For points between t_k and t_{k+1} we define the value of $\mathrm{Y}(t)$ by linear interpolation:

$$\mathrm{Y}(t) = \mathrm{Y}_k + \frac{(t - t_k)}{\delta t}\Delta\mathrm{Y}_k$$

Then, for $0 \leq t \leq s \leq \tau$, we have

$$\mathbb{E}(\mathrm{Y}(s) - \mathrm{Y}(t)) = 0.$$

This is easy to demonstrate. However, the intuitive corresponding statement $\mathrm{Var}(\mathrm{Y}(s) - \mathrm{Y}(t)) = (s - t)$ does not quite hold due to the reliance on interpolation. Nevertheless, such a relation holds in an approximate sense and in the limit of small step.

The relation $\delta t = \delta x^2$ defines a scaling of time in relation to distance. When we take limits as these two quantities go to zero in such a way that $\delta t = \delta x^2$ we refer to it as a *diffusion limit*. It is this type of random diffusion model that has the greatest relevance for experiments like those of Brown and Perrin.

In the diffusion limit the motion of the random walk can be viewed as an everywhere continuous (but almost nowhere smooth) path which is called a *Wiener process* (after Norbert Wiener).

$$\mathrm{W}(t) = \widehat{\lim}_{\delta t \to 0} \sqrt{\delta t} \sum_{k=0}^{\lfloor t/\delta t \rfloor} \mathrm{J}_k.$$

The convergence here, signified by the symbol $\widehat{\lim}$ is in the sense of distributions (see Appendix B).

As $\mathrm{W}(t)$ is the limit of a sum of independent, identically distributed random variables, it makes sense to use the Central Limit Theorem to determine its distribution. Recall (see Appendix B) that the Central Limit Theorem tells us that the sum of a set of independent, identically distributed random variables has an approximately normal distribution with mean and variance determined from the underlying parameters of the distribution. Let δt be a small finite value and $n = t/\delta t$, then, using the Central Limit Theorem, and the fact that $\mathbb{E}\mathrm{J}_k = 0$, we have, for a large number n of random steps, the approximate relation

$$\sum_{k=0}^{n-1} \sqrt{\delta t}\mathrm{J}_k \sim \mathcal{N}(0, n\delta t).$$

As we take n to infinity, δt to zero, in such a way that $n\delta t \equiv t$, we obtain the random variable $\mathrm{W}(t)$ such that

$$\mathrm{W}(t) \sim \mathcal{N}(0, t).$$

Moreover, we may easily demonstrate that

$$\mathbb{E}(\mathrm{W}(t) - \mathrm{W}(s))^2 = |t - s|. \tag{6.7}$$

For the Wiener process, we know that $\mathrm{W}(t) - \mathrm{W}(s)$ is normally distributed with mean zero and variance $|t - s|$; this does not depend on whether there is additional information available regarding its value prior to $\min(t, s)$, that is to say, the Wiener process is a *Markov process*. More generally, when we say that a stochastic process is Markovian, we mean that the probability of a future event conditioned on the current state of the process and the past history of the process is the same as the probability conditioned on the current state of the process. Let $\mathcal{A}(x)$, $\mathcal{B}(x)$ and $\mathcal{C}(x)$ be three events dependent on state variable x, and let $t_+ > t_0 > t_-$ be three times, then, for a Markov process $\mathrm{X}(t)$ we have (using the notation for conditional probability):

$$\Pr[\mathcal{A}(\mathrm{X}(t_+))|\mathcal{B}(\mathrm{X}(t_0))\,,\,\mathcal{C}(\mathrm{X}(t_-))] = \Pr[\mathcal{A}(\mathrm{X}(t_+))|\mathcal{B}(\mathrm{X}(t_0))].$$

The Markov property greatly simplifies the task of finding the joint probability distribution of the values of the stochastic process at two times.